Universidad
Carlos III de Madrid
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The state of the art research at Robotics Lab in Carlos III University of Madrid

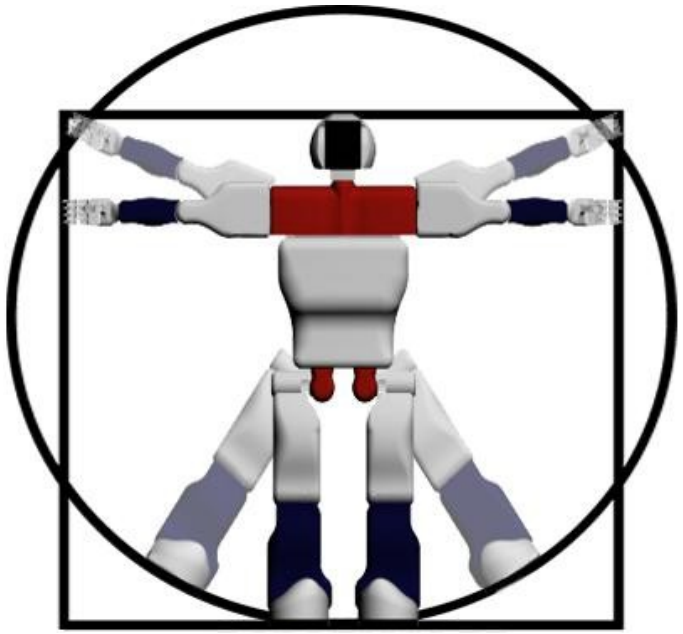
Speaker

Dr. Juan González-Gómez

Robotics Lab

Carlos III University of Madrid

(Spain)



National Robotics & Intelligent Systems Center
King Abdulaziz City for Science and Technology (KACST)

Riyadh, Saudi Arabia. March-4th-2011

Where are we?


Mapa
Tráfico
France
Svizzera
Switzerland
Slovenija
Slovenia
Hrvatska
Croatia
Bosna i Hercegovina
Bosnia and Herzegovina
Србија
Serbia
Crna Gora
Montenegro
Kosovo
Македонија
Macedonia (FYROM)
Shqipëria
Albania
Ελλάς
Greece
România
България
Bulgaria
Türkiye
Turkey
Italia
Italy
España
Spain
Portugal
Deutschland
Germany
Polska
Poland
Беларусь
Belarus
Україна
Ukraine
Suomi
Finland
Sverige
Sweden
Norge
Norway
Ísland
Iceland
Ireland
United Kingdom
Nederland
Netherlands
België
Belgium
Luxembourg
Österreich
Austria
Česko
Czech Republic
Slovensko
Slovakia
Magyarország
Hungary
Moldova
Latvija
Latvia
Lietuva
Lithuania
Eesti
Estonia
Danmark
Denmark
Andorra
Monaco
Malta
Cyprus
Gibraltar
Bay of Biscay
Tyrrhenian Sea
Mediterranean Sea
Black Sea
Caspian Sea
Algeria
Libya
Tunisia
Morocco
Mauritania
Egypt
Syria
Lebanon
Iraq
Jordan
Israel
Saudi Arabia
Kuwait
Qatar
Bahrain
Armenia
Azerbaijan
Georgia
Paris
Berlin
Roma
Madrid
Lisboa
Athina
Istanbul
Ankara
Moskva
Kyiv
Minsk
Warszawa
Wien
Praha
Budapest
Bucureşti
Beograd
Zagreb
Sofia
Skopje
Tirana
Podgorica
Sarajevo
Ljubljana
Bratislava
Chişinău
Riga
Vilnius
Tallinn
Helsinki
Stockholm
Oslo
Reykjavík
Dublin
London
Amsterdam
Brussels
Luxembourg
Bern
Vaduz
Copenhagen
Valletta
Nicosia
Alexandria
Cairo
Port Said
Damascus
Beirut
Baghdad
Amman
Jerusalem
Riyadh
Kuwait City
Doha
Manama
Abu Dhabi
Muscat
Sana'a
Tripoli
Tunis
Algiers
Rabat
Nouakchott
Khartoum
Addis Ababa
Nairobi
Kampala
Kigali
Bujumbura
Dodoma
Lusaka
Harare
Maputo
Windhoek
Gaborone
Pretoria
Johannesburg
Cape Town
Durban
Port Elizabeth
Bloemfontein
Polokwane
Nelspruit
Kimberley
Mbabane
Maseru
Lilongwe
Blantyre
Antananarivo
Port Louis
Victoria
Moroni
Mamoudzou
Dar es Salaam
Zanzibar
Mombasa
Kisumu
Eldoret
Nakuru
Thika
Machakos
Kitale
Malindi
Lamu
Garissa
Wajir
Mandera
Moyale
Isiolo
Meru
Embu
Kitui
Makueni
Kajiado
Narok
Bomet
Kericho
Nandi
Uasin Gishu
Trans Nzoia
Baringo
Laikipia
Samburu
Turkana
West Pokot
Elgeyo Marakwet
Marsabit
Tana River
Lamu
Taita Taveta
Kwale
Kilifi
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Where are we?


France
Switzerland
La Roche-sur-Yon
La Rochelle
Niort
Limoges
Clermont-Ferrand
Lyon
Genève
Annecy
Chambéry
Grenoble
Saint-Étienne
Brive-la-Gaillarde
Bordeaux
Bergerac
Bay of Biscay
Valence
Gap
Montauban
Albi
Toulouse
Nîmes
Montpellier
Aix-en-Provence
Monaco
Cannes
Marseille
Toulon
Carcassonne
Béziers
Pau
Bayonne
San Sebastián
Bilbao
Vitoria-Gasteiz
Pamplona
Andorra
Perpignan
Gijón
Santander
A Coruña
Santiago de Compostela
Lugo
Oviedo
Ponferrada
León
Pontevedra
Ourense
Vigo
Burgos
Logroño
Girona
Lleida
Mataró
Barcelona
Tarragona
Zaragoza
Valladolid
Braga
Chaves
Fafe
Paredes
Porto
Salamanca
Santa Maria Da Feira
Viseu
Guarda
Covilhã
Figueira da Foz
Coimbra
Pombal
Portugal
Cáceres
Talavera de la Reina
Toledo
Alcalá de Henares
Madrid
España
Spain
Valencia
Albacete
Torres Vedras
Vila Franca de Xira
Badajoz
Sintra
Lisboa
Alicante
Murcia
Lorca
Córdoba
Jaén
Sevilla
Huelva
Loulé
Granada
Cartagena
Almería
Málaga
Cádiz
El Ejido
Marbella
Gibraltar
Alboran Sea
Balearic Sea
Palma
Sassari
Sardegna
Cagliari
Ajaccio
Italia
Italy
Livorno
Pisa
Firenze
La Spezia
Genova
Bologna
Parma
Piacenza
Reggio nell'Emilia
Alessandria
Torino
Milano
Brescia
Verona
Vicenza
Perugia
Ravenna
Forlì
Trento
Varese
Bergamo
Novara
Chlef
Alger
Béjaia
Skikda
Annabah
Taher
Constantine
Souk Ahras
Gouvernorat de Tunis
Kélibia
Blida
Bordj Bou Arreridj
Sétif
Batna
Khenchela
Thala
Kairouan
Sousse
Monastir
Relizane
Mouaskar
Ain Oussara
Ouahran
Tiaret
Tanger
Tétouan

Welcome to Madrid









Prado Museum







Real Madrid Soccer team











Carlos III University of Madrid

- Founded in 1989
- 18.500 students
- 40 Bachelor Degrees
- 27 Departments: Bioengineering, Aerospace, Computer science, Electronics, Mathematics, Physics, Automation...



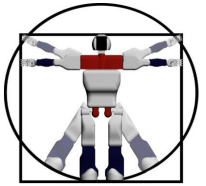
Department of System Engineering & Automation

Laboratorio de Sistemas Inteligentes





Robotics Lab



Universidad Carlos III de Madrid
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Robotics Lab



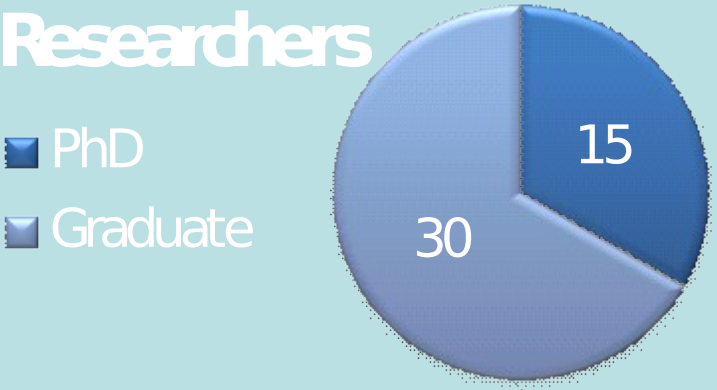








Researchers



Pioneers in robotics







A multidisciplinary team

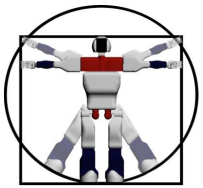






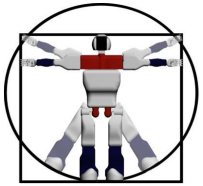






People

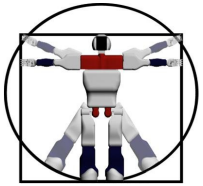
- Professors (permanent): 3 full prof. and 4 associate prof.
- Visiting professors: 5
- Assistant professors: 4
- Ph.D students: 24
- Part time Prof: 2
- Lab technicians and admin: 5
- **Total: 47**
- International and interdisciplinary group
  - Norway, France, Italy, Algeria, India, Venezuela, Colombia, Spain...
- http://roboticslab.uc3m.es



History

Pioneer in Spain's Robotics

- Developed the 1st mobile robot in Spain
- Developed the 1st industrial robot manufactured in Spain (Scara type)
- Developed some mobile manipulators: OTILIO, Manfred
- Developed the 1st humanoid in Spaim: RH0
- Developed the 1st mobile with human-interaction capabilities in Spain: Maggie



Robot types and applications

- Aerospace
- Assistive Robots
- Climbing Robots
- Humanoids
- Mobile manipulators
- Social robots
- Mobile robotics
- Robots in construction
- Dexterous manipulation
- Vision for manipulating
- SMA actuators
- Robots in education
- Modular robots

Aerospace

• The activity of the Robotics Lab in relation to Aerospace is focused on two completely different areas:

- Visual relative navigation of autonomous satellites
- Simulation of some general systems of the Eurofighter Aircraft





Assistive Robots

- Robot assistant for disable nad elderly people (MATS), general interactive intelligent assistant (PRA), ASIBOT
- Therapeutic Rehabilitation using Robot Manipulators (Colaboration with Guelph University, Canada)

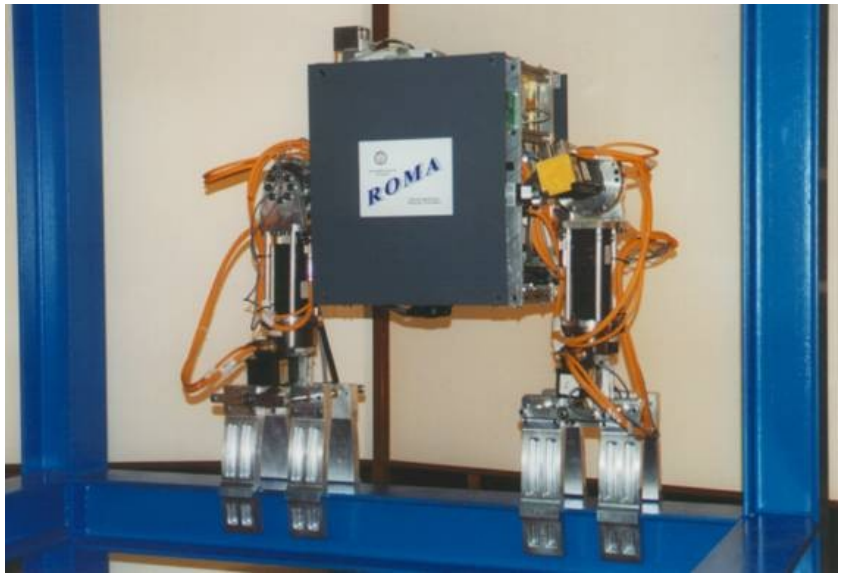


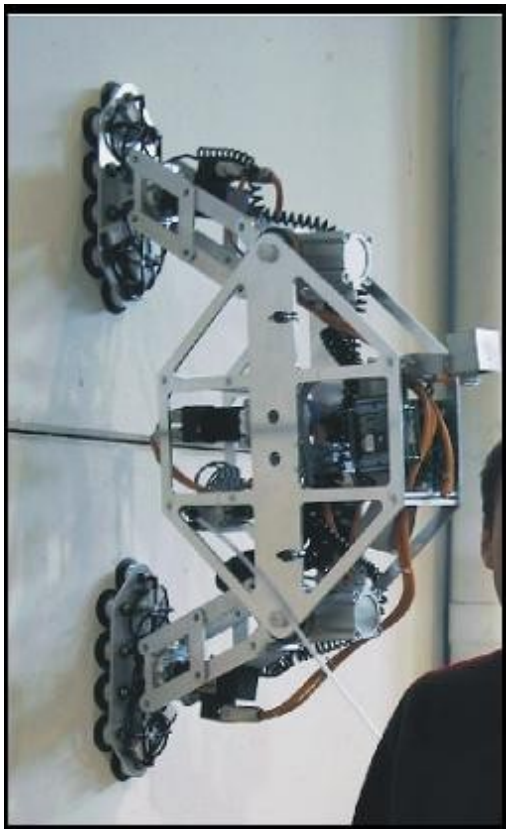




Climbing robots

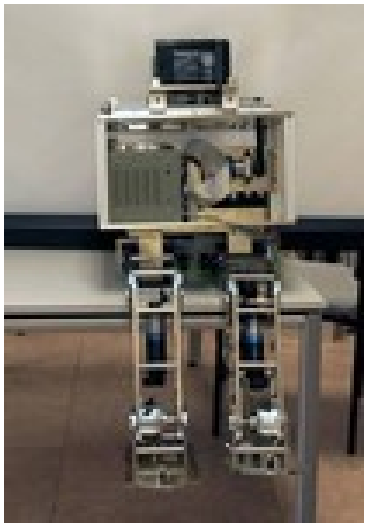
- Research in the field of climbing robots started in 1995.
- Since this date several robots had been developed:
- Roma 1 for inspection steel-based structures,
- Roma 2 for travel along concrete, wood, etc., and
- Mats robot able of moving in domestic environments





Humanoids

- Humanoid robots program of the RoboticsLab started on 2001 by developing the 7 DOF bipedal robot Leroy





- Since 2002: Rh-0 project, and later Rh-1 one, have the objective of developing a 21 DOF full size robot.

- The lab uses purchased full systems for research purposes (Hoap, from Fujitsu)





Mobile manipulators

- The objective of this research line is to develop advanced mobile manipulators able to operate in human environments. Manfred integrates perception, control, local and global planning in order to cope with a wide spectrum of typical tasks







Personal/social robots

- Current robots are present mainly in factories. Future robots will be also in our homes as personal companions to entertain and help us in our daily life

- It is a platform to research in multimodal interaction and emotions in robotics.







Mobile Robotics

- Mobile Robotics is one of the most active research fields of the Robotics Lab. Our group works are focused in control architectures, topological navigation, multimodal interaction. All in structured and open environments





Robots in construction

- The R&D activities of the RoboticsLab in the field of robotics & automation in construction started in early 90s.
- Several industrial and EU projects in automation of the pre-fab parts, modular building design & erection had been done



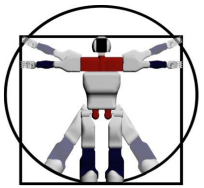






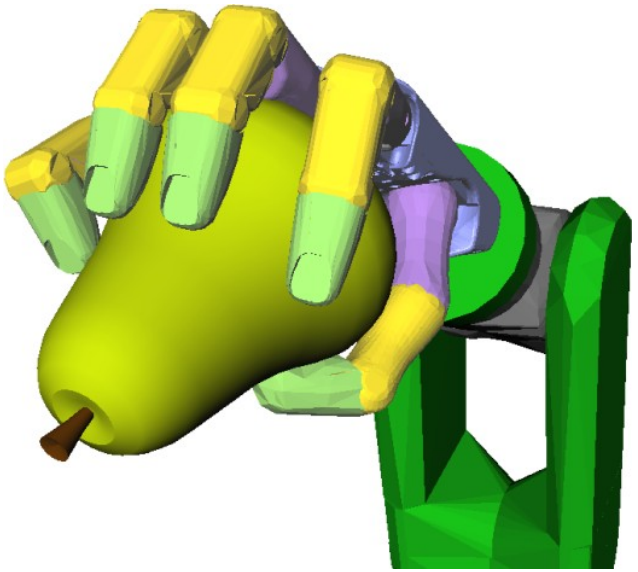
Professor's Mohamed team

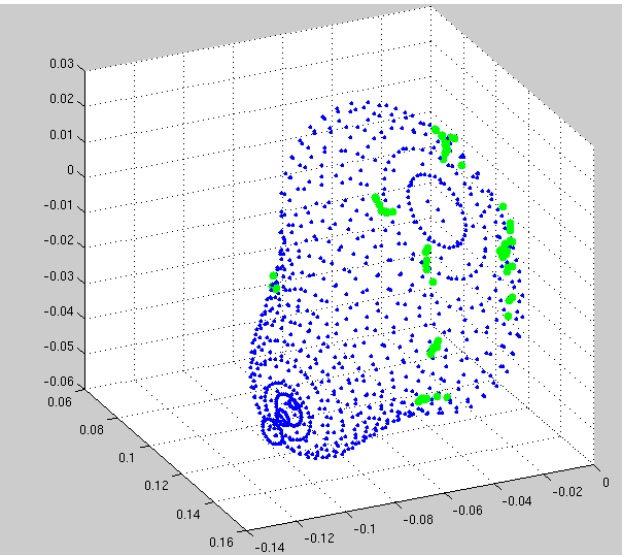




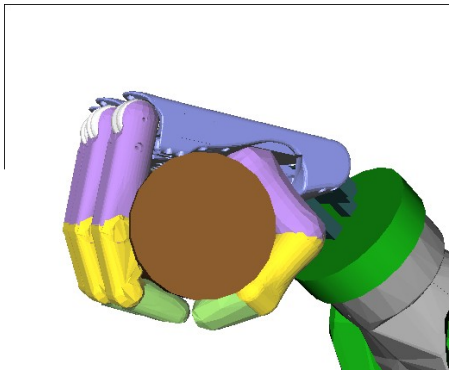
Dexterous manipulation

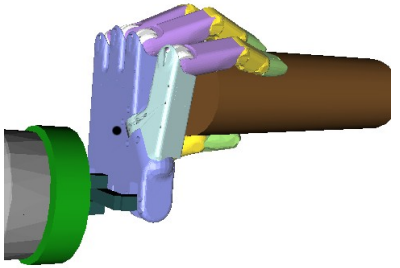
- Objects pose estimation using tactile data



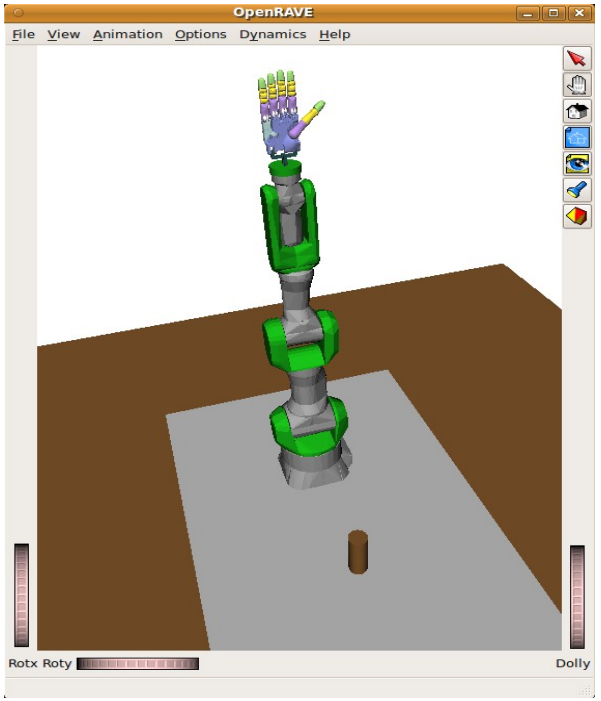


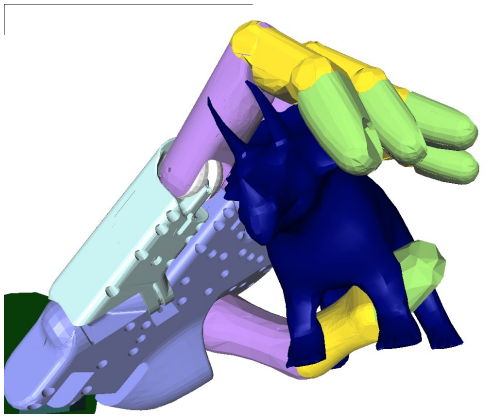
Dexterous manipulation

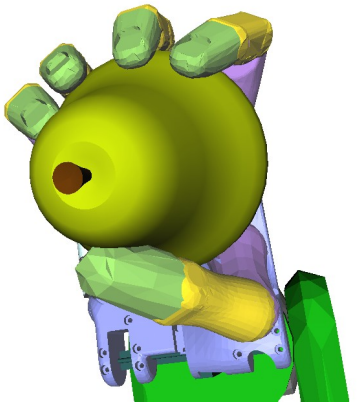






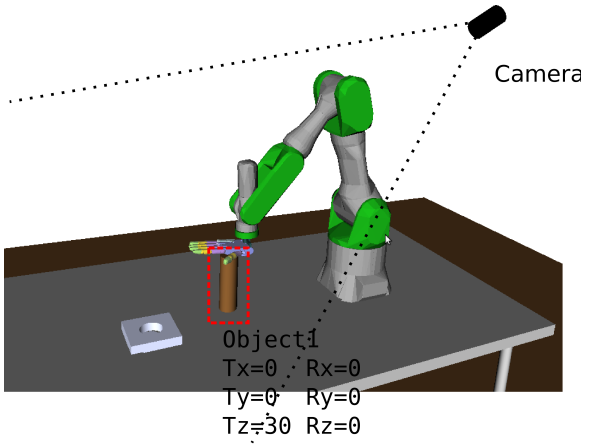






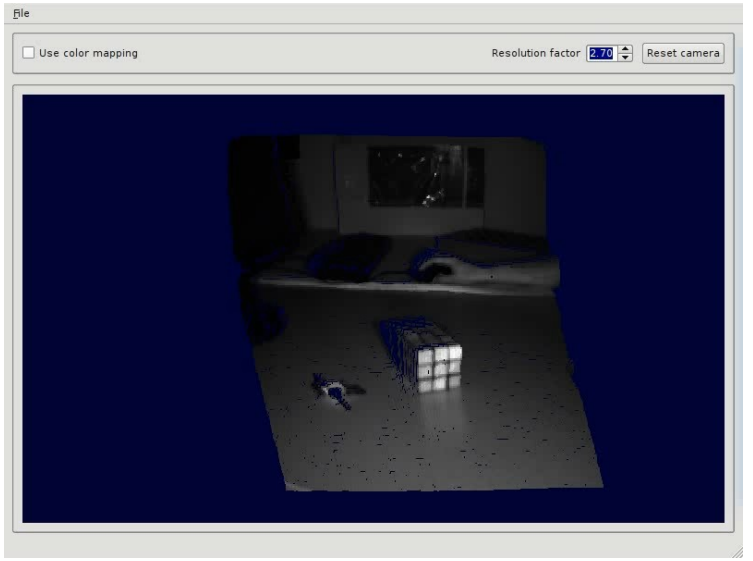
Vision for manipulating

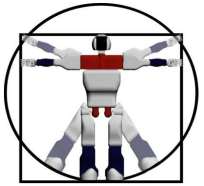
- **Three main aspects**
  - Object 3D model acquisition
  - Object detection and recognition
  - Pose estimation and tracking
- Rely on TOF cameras:







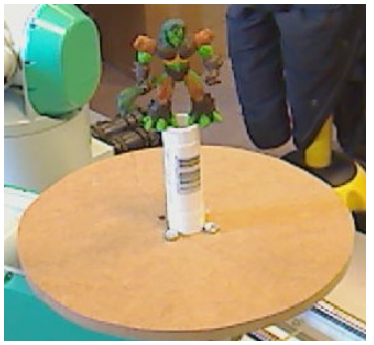




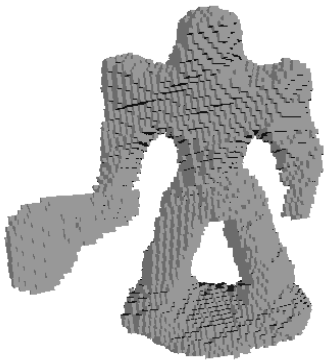
Vision for manipulation

- Model acquisition:

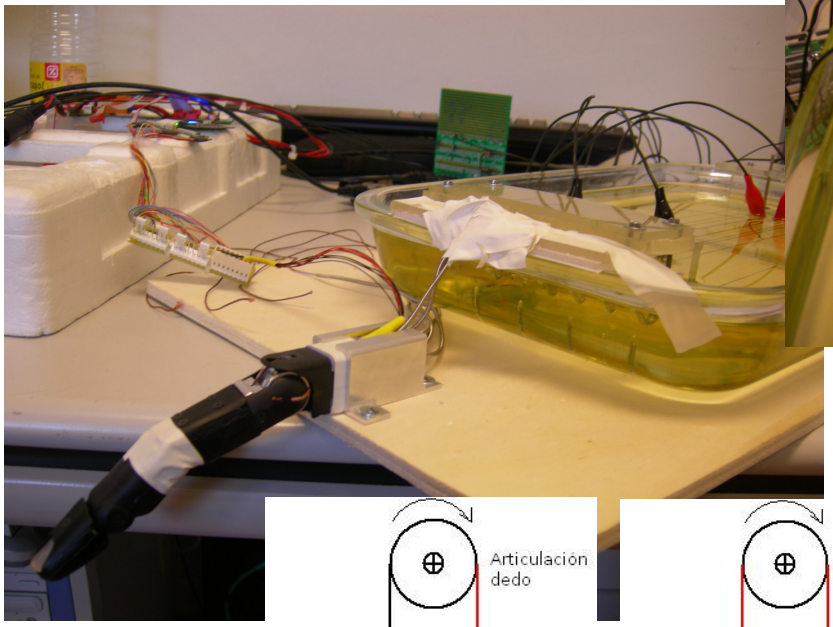


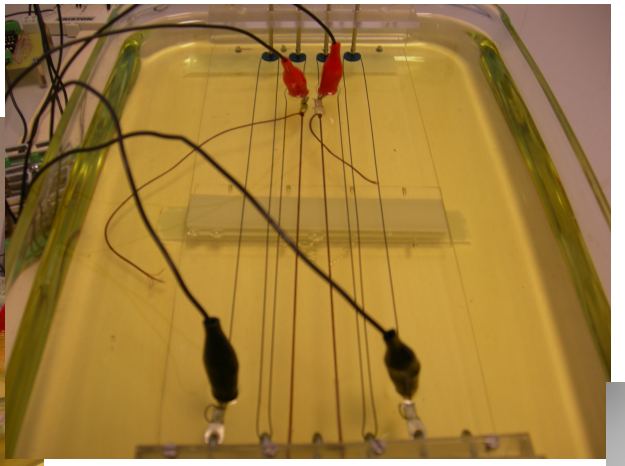


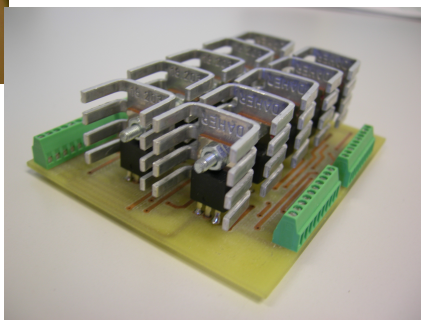


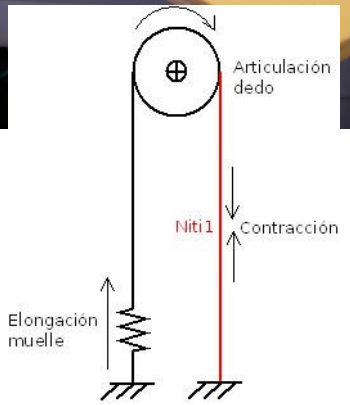


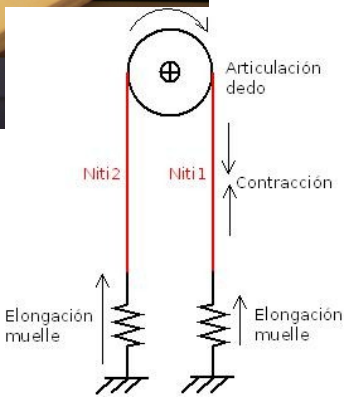
SMA based actuators for robot finger control







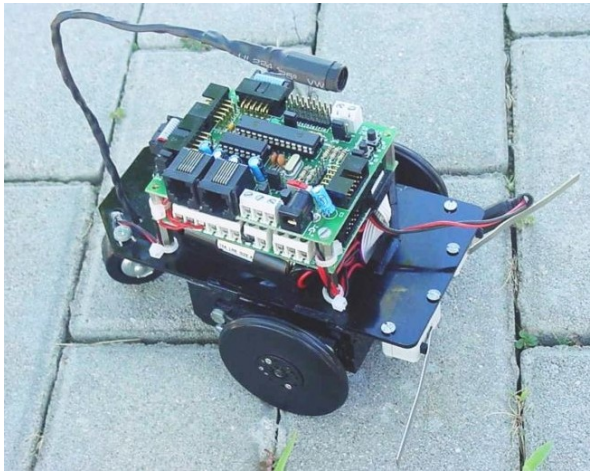


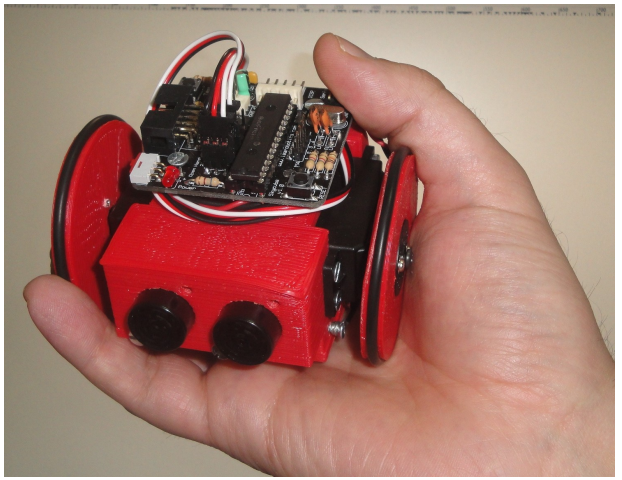




Mobile robots for edutainment

- **Skybot robot**:
  - 4 Infrarred sensors
  - 2 contact sensors
  - 2 light sensors
- **Mini-Skybot robot**:
  - 3D-printable
  - Open source

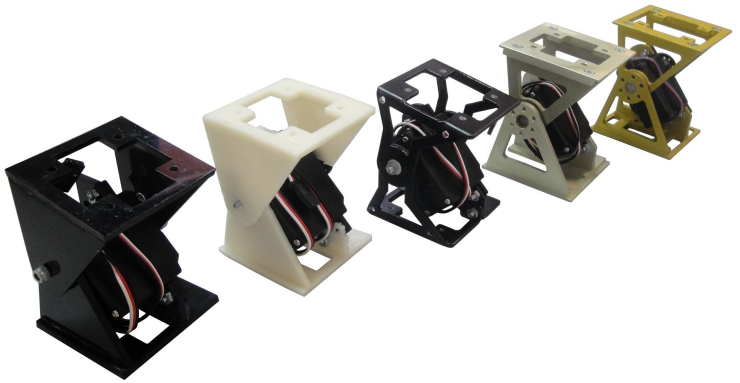




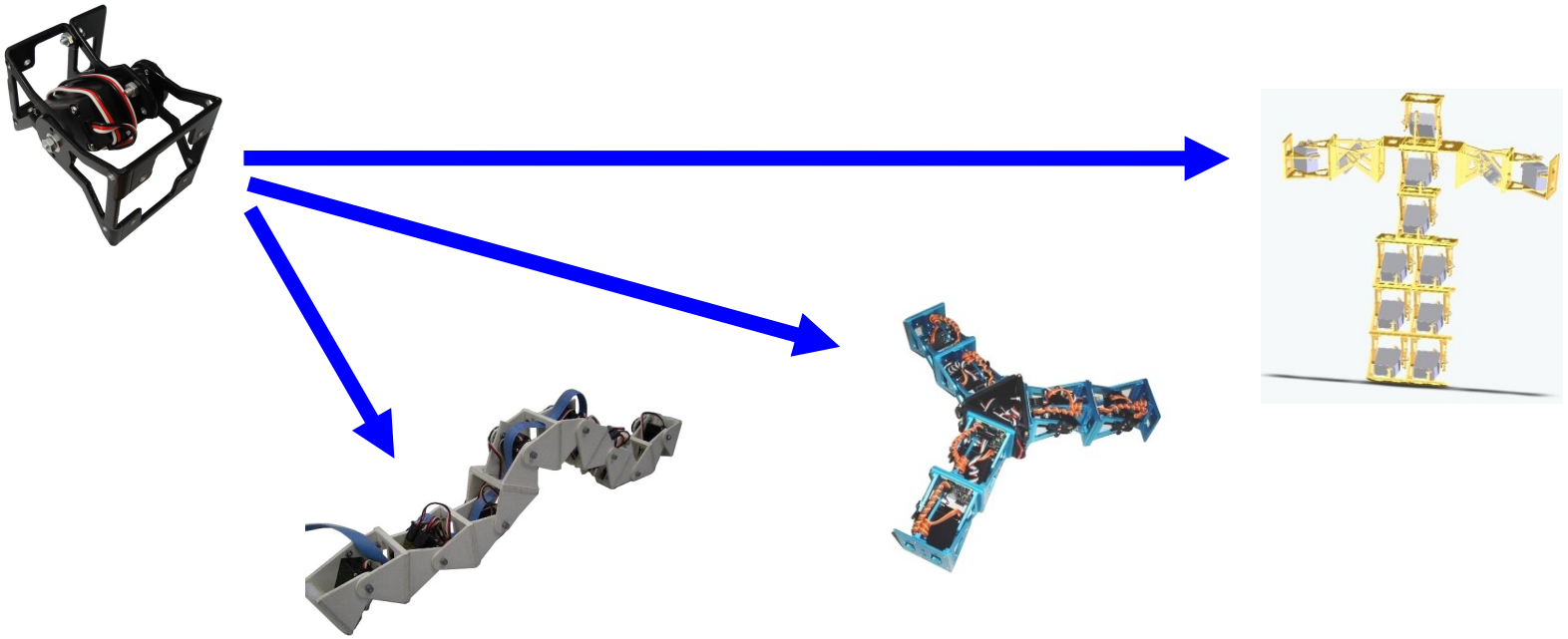


Modular robots

- Module design:



- Multiple configurations:



The state of the art research at Robotics Lab in Carlos III University of Madrid

Speaker

Dr. Juan González-Gómez

Robotics Lab

Carlos III University of Madrid

(Spain)

